

17.1 The current in a toaster is 12A. (a) how much charge passes through the toaster in 1 minute? (b) How many electrons pass through the device in 1 minute?

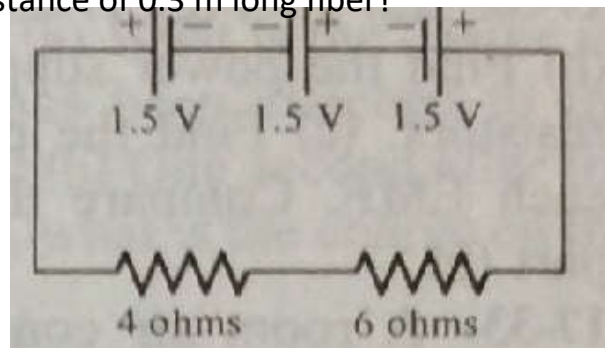
17.5 Silver has 5.8×10^{28} free electrons per cubic meter. If the current in a silver wire is 10A, and the wire has a radius of 10^{-3} m, what is the drift velocity of the electrons?

17.10 A 10-A current in a wire results in a potential difference of 2-V across its ends. If it is an ohmic conductor, what current will produce a 6V potential?

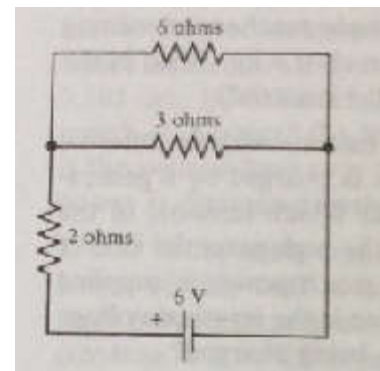
17.11 The thinnest copper wire normally manufactured has a radius of 4×10^{-5} m. Find the resistance of a 10-m long segment.

17.12 A nerve fiber (axon) may be approximated as long cylinder. If its diameter is 10^{-5} m and its resistivity $2\text{-}\Omega\text{m}$, what is the resistance of 0.3 m long fiber?

17.25 (a) Find the current in the circuit shown (b) Find the potential difference across each element.



17.45 For the circuit shown in the Figure, find (a) the current through the $2\ \Omega$ resistance, (b) the voltage difference across the $3\ \Omega$ resistance, (c) the current through the $3\ \Omega$ resistance.



17.34 A heater uses 1400W of power when connected to a 120 V line, (a) what is its resistance? (b) what current does it draw?

17.47 A Galvanometer has a resistance of $10\ \Omega$ and gives full scale deflection with current of 10^{-3} A. How one can convert it into a voltmeter with a 0.1V range?

17.50 A $1000\ \Omega$ resistor and a 10^{-5} F capacitor are connected in series to a 100V. (a) Find the time constant. (b) what is the final charge on the capacitor?

17.51 A 10^{-4} F capacitor is discharged through a 100Ω resistor. How long does it take for the charge on the capacitor to drop to $1/e^2$ times its original value?